Bony fishes

PHYLUM	Chordata
CLASS	Osteichthyes
ORDERS	67
FAMILIES	481
SPECIES	About 31,000

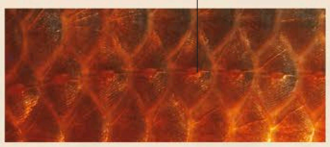
CLASSIFICATION NOTE

Bony fishes account for more species than any other class of vertebrates. They are split into 2 subclasses: fleshy-finned fishes (Sarcopterygii) have fins that look like flippers, joined to the body by fleshy lobes; ray-finned fishes (Actinopterygii) have fins supported by bony fin rays. Ray-finned fishes are by far the larger group and are divided into 9 superorders. Classification of bony fishes is in a state of constant change as new species are discovered and zoologists learn more about the relationships between them.

Bony fishes form overwhelmingly the largest and most varied group of fishes, accounting for more than 9 out of 10 species. Of the 8 classes of fishes, they evolved most recently and are usually regarded as the most advanced. They exhibit an astonishing variety of size, color, and shape. All bony fishes have a light but strong internal skeleton, made entirely or partly of bone, which supports flexible fins that enable the fish to control its movements with precision. Most also have a gas-filled swim bladder that allows them to adjust their buoyancy within narrow limits. Bony fishes are found in almost all aquatic habitats, including marshes, lakes, rivers, coasts, reefs, and deep oceans. Many have adapted to extreme conditions, and species of bony fishes occur in high-altitude lakes, polar coasts, hot springs, high-salinity ponds, acidic streams, and low-oxygen swamps.

Senses

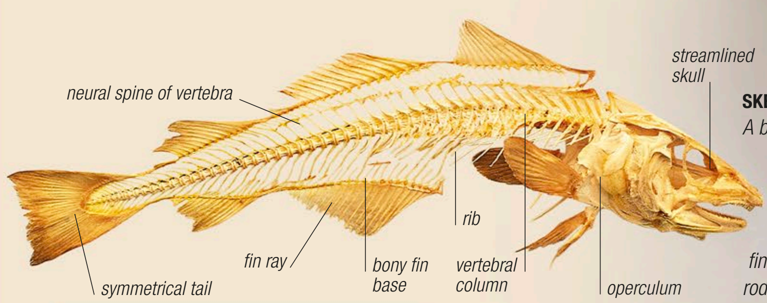
Most bony fishes have keen senses of vision and hearing. Both of these senses are used in communication and social interaction. They are also thought to contribute to bony fishes' well-developed ability to school. The eyes are generally set on the sides of the head, giving each fish a wide field of view. Rods and cones in the retina give them good color vision, and many species have bright coloration, which helps them attract mates and defend territory. Sound is an effective means of underwater communication. Some species of bony fishes can produce sounds, either using their swim bladder or by rubbing parts of their body together.



LATERAL LINE
The lateral-line system, which is used to detect movement vibrations, is highly developed in bony fishes. This is thought to be another reason why they can move effectively in large schools.

Anatomy

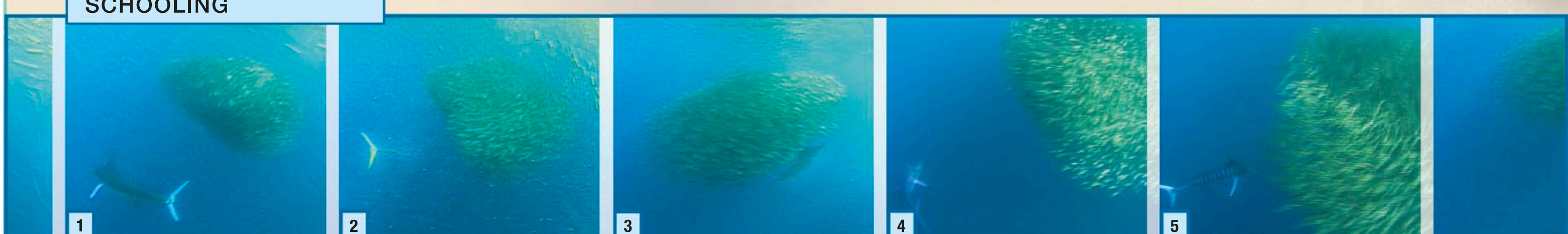
The skeleton of a bony fish consists of 3 main units: the skull, backbone, and fin skeleton (see below). The gills of bony fishes are paired and located behind the head and below the cranium. Unlike in other groups of fishes, the gill openings are covered by a bony flap, called the operculum, and the lower gill chamber contains bony supports, called branchiostegal rays. These 2 structures enable the fish to take water into its mouth and then pump it over its gills: the branchiostegal rays help open the mouth and regulate intake, while the operculum acts as a seal over the gills to control the outflow. In this way, a bony fish can respire while stationary. Most bony fishes have light, flexible, cycloid or ctenoid scales (see p.468), usually covered by a thin layer of skin that secretes mucus. The mucus repels parasites and disease-causing organisms; in some species, it also prevents moisture loss. Other bony fishes have large, protective scales or no scales at all (although their skin still produces mucus).



SKELETON
A bony fish's skull encases the brain and supports the jaws and gill arches. Teeth may be found in the jaws, in the throat, on the roof of the mouth, or on the tongue. The backbone, consisting of articulated vertebrae that are linked to the ribs in the abdomen, provides support for the body. Each fin consists of a bony base embedded in the body, from which rodlike structures extend to form the outer fin.

SWIMMING
Bony fishes, such as these French grunts, can maneuver precisely. Each fin ray is controlled by a separate set of muscles so that it can move independently of the others. The symmetrical tail typical of bony fishes is an efficient tool for propelling the fish through water.

SCHOOLING



1 ATTRACTING PREDATORS
A marlin has isolated a shoal of yellowtail horse mackerel between itself and the surface.

2 STAYING TOGETHER
As the marlin swims toward the shoal, the mackerel repeatedly change direction, staying together.

3 CLOSING RANKS
While the marlin circles below them, the mackerel pack themselves even more closely together.

4 USING THE LIGHT
The marlin keeps the shoal near the surface and picks off fish from below.

5 ON THE ATTACK
The marlin repeatedly attacks. If the fish are forced to split, they will try to rejoin the largest group.